Activation functions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

TEACHER GUIDE · 1 OF 9

Prepare the purpose, prerequisites and materials

> Prepare Activation functions: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Trace identity **ReLU** and **sigmoid** on a numerical score

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Bias terms and decision boundaries

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate a weighted score with a bias term

Reference example, not the only acceptable wording: Teacher explanation: Weighted sum 10 - 4 = 6; bias -3; score 3; **prediction** 1. Add the bias once after summing.

Representing inputs as vectors

Without opening the old lesson, explain one key idea from Representing **input**s as vectors. Give an example and a mistake that the idea helps you avoid.

Look for: Encode a record using a fixed feature order and units

Reference example, not the only acceptable wording: Teacher explanation: A [4, 2], B [2, 4], C [3, 3]. Each has dimension 2. The table has 3 rows and 2 columns.

For the compact alternative schedule, use two 45-minute sessions to compare **activation function**s before network **training**. Students should trace z, activation a and any later **output** separately. **Sigmoid** arithmetic may use the supplied approximations.

Focus on the change in function shape and the distinction between an activation, class decision and probability claim.

Outcomes

Trace identity **ReLU** and **sigmoid** on a numerical score

Explain how a nonlinear activation changes a computation

Distinguish an activation **output** from a final class or verified probability

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

The score is zero and the class is 1 because equality is included.

End goal for the pictorial case

By the end, I can explain **activation function**s using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model Activation functions using the supplied case, labelled mechanism and reasoning. Make

> the scope of each conclusion explicit.

Explain Activation functions · 1

First compute an affine score z = w dot x + b. An activation then returns a = f(z). In a network this value may become an **input** to another layer; it is not automatically a class **prediction**.

We will compare three functions using the same incoming scores so that only the activation changes.

Identity returns z unchanged. **ReLU**, pronounced ree-loo, returns the larger of zero and z. Thus **ReLU**(-2) = 0, **ReLU**(0) = 0 and **ReLU**(2) = 2. Positive scores keep their magnitude; **ReLU** does not turn every positive value into 1.

These are fixed arithmetic functions. Selecting a function or changing its **input** is not itself a **training** procedure. Later weeks will introduce losses and parameter updates.

Sigmoid is 1 / (1 + exp(-z)). Here exp(t) means e raised to power t, where e is approximately 2.718. A calculator or the supplied Python code can evaluate it; no calculus is required this week.

For z = -2, 0 and 2, **sigmoid output**s are approximately 0.1192, 0.5 and 0.8808.

Explain Activation functions · 2

For every finite mathematical **input**, **sigmoid** is strictly between 0 and 1, though computer rounding can produce an endpoint for extreme scores.

Negative **input**s produce values below 0.5, zero gives 0.5, and positive **input**s give values above 0.5. A class rule still needs to be stated: if class 1 means **sigmoid**(z) >= 0.5, these three **prediction**s are 0, 1 and 1.

The two algebraically equivalent branches avoid taking exp of a large positive number. A value within [0, 1] is not automatically a reliable probability: its interpretation depends on the fitted task and evaluation **evidence**.

Term	Meaning
Activation function	A function applied to a computed score
ReLU	The function max(0, z), retaining positive scores and replacing negative scores with zero

Term	Meaning
Sigmoid	The function 1 / (1 + exp(-z)), mapping finite real scores into the interval between 0 and 1
Nonlinear function	A function that cannot be represented by one fixed slope and offset over its entire domain

Three functions receive the same score

SAME INCOMING SCORE	APPLY ONE FUNCTION	ACTIVATION OUTPUT
z is -2	Identity of z	-2
z is -2	ReLU of z	0
z is -2	Sigmoid of z	About 0.1192

Each row starts again from the same score -2 and applies one function. The functions are alternatives here, not successive steps. These outputs are activation values, not class labels.

Explain Activation functions · 3

Guided: identity **output**s -1 and 1; **ReLU output**s 0 and 1. **Sigmoid** class **prediction**s under >= 0.5 are 0 and 1. Exit: the computation establishes an activation of about 0.8808 for score 2.

It does not establish calibrated probabilities, correctness on this individual case or reliability on future populations.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Three functions receive the same score

SAME INCOMING SCORE	APPLY ONE FUNCTION	ACTIVATION OUTPUT
z is -2	Identity of z	-2
z is -2	ReLU of z	0
z is -2	Sigmoid of z	About 0.1192

Each row starts again from the same score -2 and applies one function. The functions are alternatives here, not successive steps. These outputs are activation values, not class labels.

For z = -1 and z = 1, give identity and ReLU outputs. Using sigmoid(-1) = 0.2689 and sigmoid(1) = 0.7311, give classes under the rule a >= 0.5.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Activation functions

See what **ReLU** removes before the next weighted step.

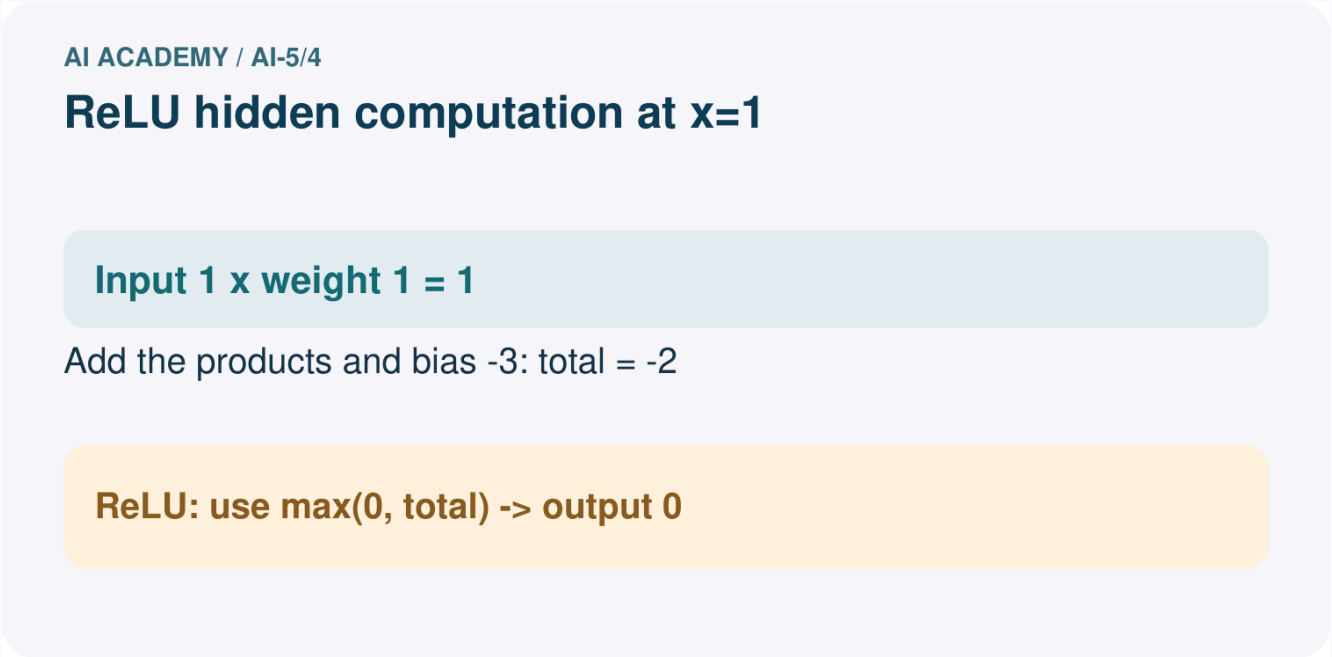
The facts we will use

Supplied facts

Toy network z=x-3; h=max(0, z); y=3h+2. **Input**s x=1, 3, 4.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/4

Activation functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute preactivation

z values are -2, 0, 1.



2. Apply ReLU

h values are 0, 0, 1; the negative value is replaced by0.



3. Compute output

y values are 2, 2, 5; x = 1 and x = 3 collapse to the same output.

CHANGE ONE THING

Replace ReLU with identity.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute preactivation	z values are -2, 0, 1.
2. Apply ReLU	h values are 0, 0, 1; the negative value is replaced by0.
3. Compute output	y values are 2, 2, 5; x = 1 and x = 3 collapse to the same output.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

ReLU means keep a positive value and replace a negative one with zero. The **output** layer receives that transformed value, not the original **input**.

Inputs one and three therefore both deliver zero to the final layer and produce two. Replacing **ReLU** with identity changes the procedure: negative intermediate values now remain available to the next layer.

What this example does not establish

Keep the stated limit

ReLU is one activation choice. Clipping negatives does not itself train the network or prove

useful representation.

Change one thing in the pictured case

Change one thing

Change: Replace **ReLU** with identity.

Predicted result

Prediction: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

Why the result follows

Reason: Identity retains the negative intermediate value instead of clipping it.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Activation functions, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite boundary example. Introduce z as the score and a as the transformed value.
7 to 18 minutes	Trace identity and ReLU for -2, 0 and 2. Complete W1, checking that positive values keep magnitude.
18 to 32 minutes	Introduce the sigmoid formula and supplied values. Trace or run the stable code; complete W2 using the explicit equality policy.
32 to 45 minutes	Compare the visual rows and complete the guided -1 and 1 case. Keep activation values separate from classes.
Session 2 start to 8 minutes	Retrieve ReLU at one negative and one positive input. Ask whether one line covers both sides.
8 to 20 minutes	Complete W3 and trace the two worked chains at x = -1 and x = 1. Show the identity-chain simplification by substitution.
20 to 35 minutes	Collect W4 independently, requiring a three-stage table. Do not award full credit for final y values without the intermediate trace.
35 to 45 minutes	Complete W5 and the exit response. Use the fresh retry after feedback; connect to composing layers next week.

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use a number line and a three-column z, a, y table. Supply **sigmoid** values rather than requiring exponent calculations from memory.

Extension: use several **input**s to verify that **sigmoid**(z) and **sigmoid**(-z) sum to 1 within rounding error, while noting that this symmetry is not **evidence** of calibration.

Reduce help deliberately

Completion practice

Completion task: Use the lesson **sigmoid** values for -2, 0, 2. Predict classes under a >= 0.5, then explain the zero-score case.

Reasoning to listen for

Teacher check: Teacher explanation: Classes are 0, 1, 1. **Sigmoid**(0) equals 0.5 and the declared >= comparison includes equality.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Activation function**s. Proposed explanation: "**ReLU** returns a class label". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Positive scores retain magnitude, so **ReLU**(2) is 2. A final class rule is separate.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which original **input**s cannot be distinguished by this network's final **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

x=1 and x=3 both give h = 0 and y = 2. After **ReLU**, the downstream step receives identical values for those **input**s. Changed-case check: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

Identity retains the negative intermediate value instead of clipping it.

Changed-case reasoning: h becomes -2, 0, 1 and **output**s become -4, 2, 5. Identity retains the negative intermediate value instead of clipping it.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the investigation and diagnose misconceptions

> Run the Activation functions investigation and use the first unsupported step to choose a teaching response.

ReLU returns a class label: Positive scores retain magnitude, so **ReLU**(2) is 2. A final class rule is separate.

A bounded number is a calibrated probability: A range restriction comes from the formula. Calibration needs appropriate **evidence**.

Nonlinearity guarantees a better result: It changes which functions can be represented; data, **fitting** and valid evaluation still determine model quality.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Activation functions questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: identity **output**s -1 and 1; **ReLU output**s 0 and 1. **Sigmoid** class **prediction**s under >= 0.5 are 0 and 1. Exit: the computation establishes an activation of about 0.8808 for score 2.

It does not establish calibrated probabilities, correctness on this individual case or reliability on future populations.

W1 Trace three activations

Workbook question · W1

For scores -2, 0, 2, list identity and **ReLU output**s. Explain why **ReLU**(2) is not 1.

Teacher reasoning · W1

Teacher explanation: Identity gives -2, 0, 2; **ReLU** gives 0, 0, 2. **ReLU** retains the positive score itself; it is not a step classifier.

W2 Apply a declared class rule

Workbook question · W2

Use the lesson **sigmoid** values for -2, 0, 2. Predict classes under a >= 0.5, then explain the zero-score case.

Teacher reasoning · W2

Teacher explanation: Classes are 0, 1, 1. **Sigmoid**(0) equals 0.5 and the declared >= comparison includes equality.

W3 Check the midpoint

Workbook question · W3

Compare **ReLU** at **input**s -2, 0, 2. Give the average of the two endpoint **output**s and the **output** at the midpoint; explain what this shows.

Teacher reasoning · W3

Teacher explanation: Endpoint **output**s 0 and 2 average to 1; the midpoint **input** 0 produces 0. The mismatch shows **ReLU** is not one affine function over these **input**s.

W4 Trace a new chain independently

Workbook question · W4

Let z = x - 2, a = **ReLU**(z), y = 2a + 1. For x = 0, 2, 5, record every z, a and y. Then replace **ReLU** with identity and give all three y values. Which **input**s have the same **ReLU output**, and why?

Teacher reasoning · W4

Teacher explanation: For x = 0, 2, 5: z = -2, 0, 3; a = 0, 0, 3; y = 1, 1, 7. Identity gives y = -3, 1, 7. **Input**s 0 and 2 both give a = 0 because **ReLU** maps negative and zero scores to zero.

W5 Limit a probability claim

Workbook question · W5

A model returns a **sigmoid** value 0.9 on one case. Does the numerical range alone establish a calibrated 90 percent probability? Explain what needs checking.

Teacher reasoning · W5

Teacher explanation: No. It establishes only the computed activation value.

Probability interpretation requires a suitable fitted model and evaluation, including checking predicted probabilities against outcome frequencies on relevant separate data; one value does not establish calibration.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

Assess W4 for z values -2, 0, 3, **ReLU** values 0, 0, 3, final **output**s 1, 1, 7 and identity **output**s -3, 1, 7. Fresh retry: z = 2x - 3, a = **ReLU**(z), y = a - 1 at x = 1, 1.5, 3.

The triples (z, a, y) are (-1, 0, -1), (0, 0, -1), (3, 3, 2). Require an explanation of which distinctions were erased.

Assess independent transfer

Independent question · withhold the key

Selected case: Let z = x - 2, a = **ReLU**(z), y = 2a + 1. For x = 0, 2, 5, record every z, a and y. Then replace **ReLU** with identity and give all three y values. Which **input**s have the same **ReLU output**, and why?

Selected case answer

Teacher key · Selected case answer

Teacher explanation: For x = 0, 2, 5: z = -2, 0, 3; a = 0, 0, 3; y = 1, 1, 7. Identity gives y = -3, 1, 7. **Input**s 0 and 2 both give a = 0 because **ReLU** maps negative and zero scores to zero.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Activation functions to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

Add a labelled **activation function**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace identity **relu** and **sigmoid** on a numerical score.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use a number line and a three-column z, a, y table. Supply **sigmoid** values rather than requiring exponent calculations from memory.

Extension: use several **input**s to verify that **sigmoid**(z) and **sigmoid**(-z) sum to 1 within rounding error, while noting that this symmetry is not **evidence** of calibration.

Week 5 composes layers and traces how activations become **input**s to the next computation.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/5, ai-5/7, ai-5/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Activation functions lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Trace identity **ReLU** and **sigmoid** on a numerical score

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Use the lesson **sigmoid** values for -2, 0, 2. Predict classes under a >= 0.5, then explain the zero-score case.

Supported practice answer

Reasoned practice answer: Teacher explanation: Classes are 0, 1, 1. **Sigmoid**(0) equals 0.5 and the declared >= comparison includes equality.

Check every learner before discussion: Use the worked case for **Activation function**s. Proposed explanation: "**ReLU** returns a class label". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Positive scores retain magnitude, so **ReLU**(2) is 2. A final class rule is separate.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Replace **ReLU** with identity. Require a **prediction** and a reason before comparing. Expected result: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

Reason: Identity retains the negative intermediate value instead of clipping it.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

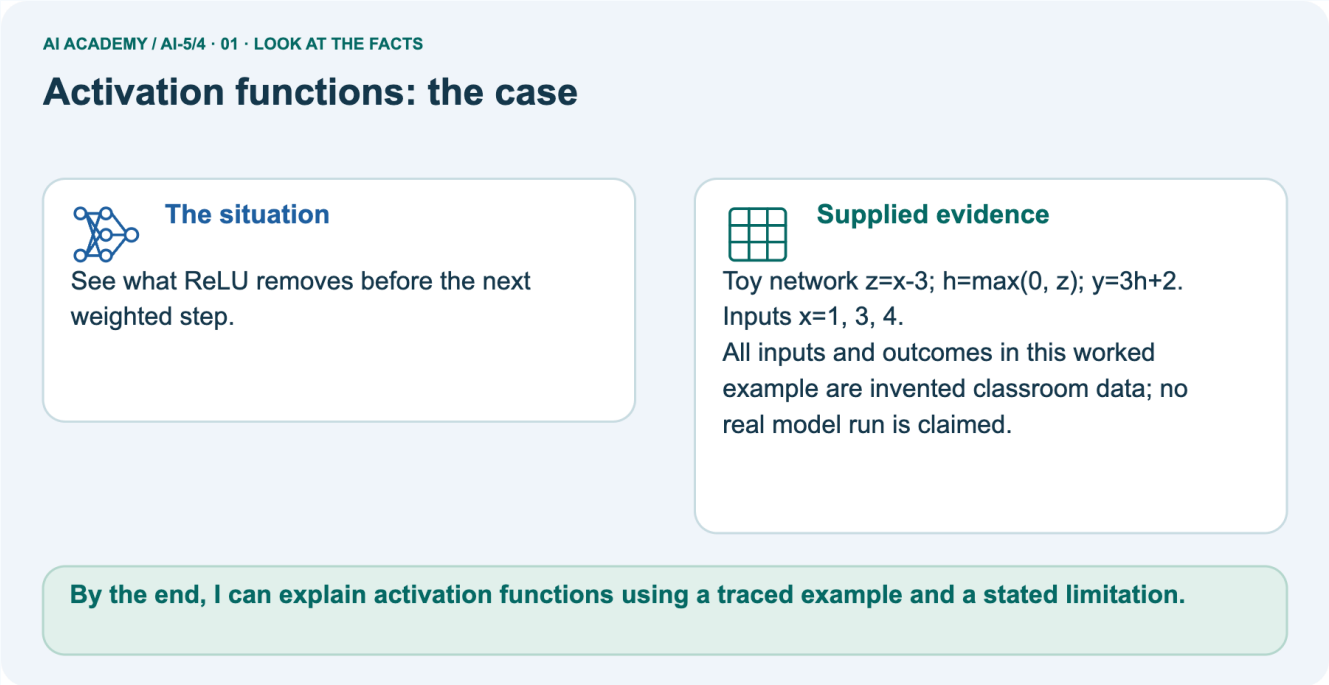
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Activation functions graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Activation functions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

See what **ReLU** removes before the next weighted step. Supplied facts: Toy network z=x-3; h=max(0, z); y=3h+2. **Input**s x=1, 3, 4.

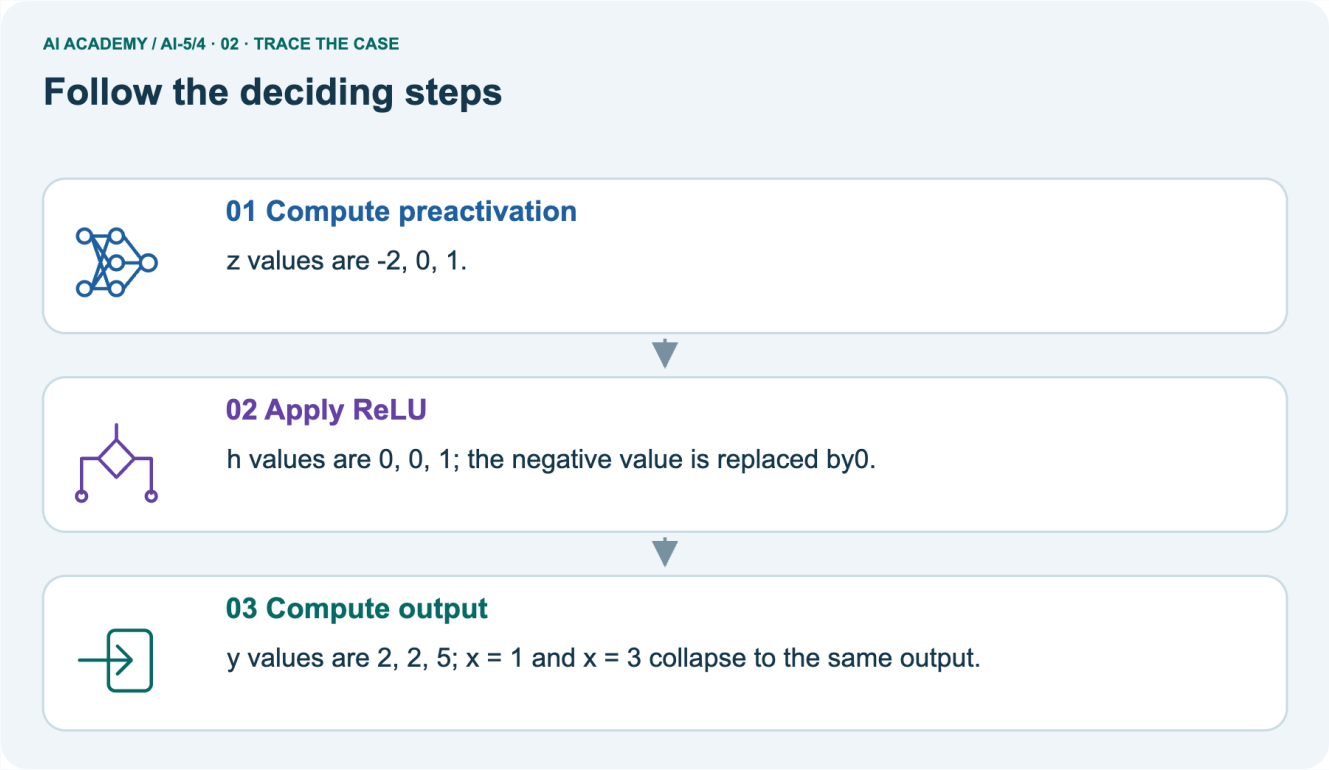
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

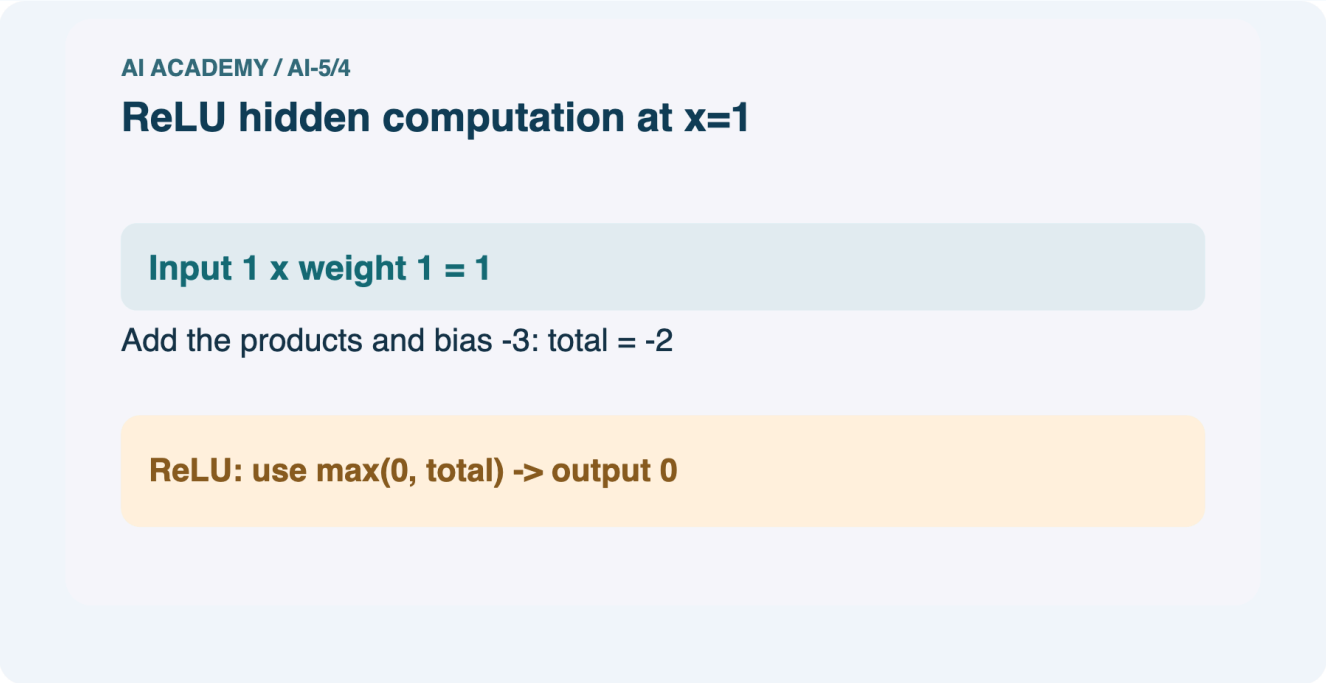
Compute preactivation: z values are -2, 0, 1. → Apply **ReLU**: h values are 0, 0, 1; the negative value is replaced by0. → Compute **output**: y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. ReLU hidden computation at x=1



Follow the labelled picture with the matching explanation. Open the vector diagram

ReLU hidden computation at x=1. **Input**s: [1]; weights: [1]; bias: -3; activation: **relu**; **output**: 0.

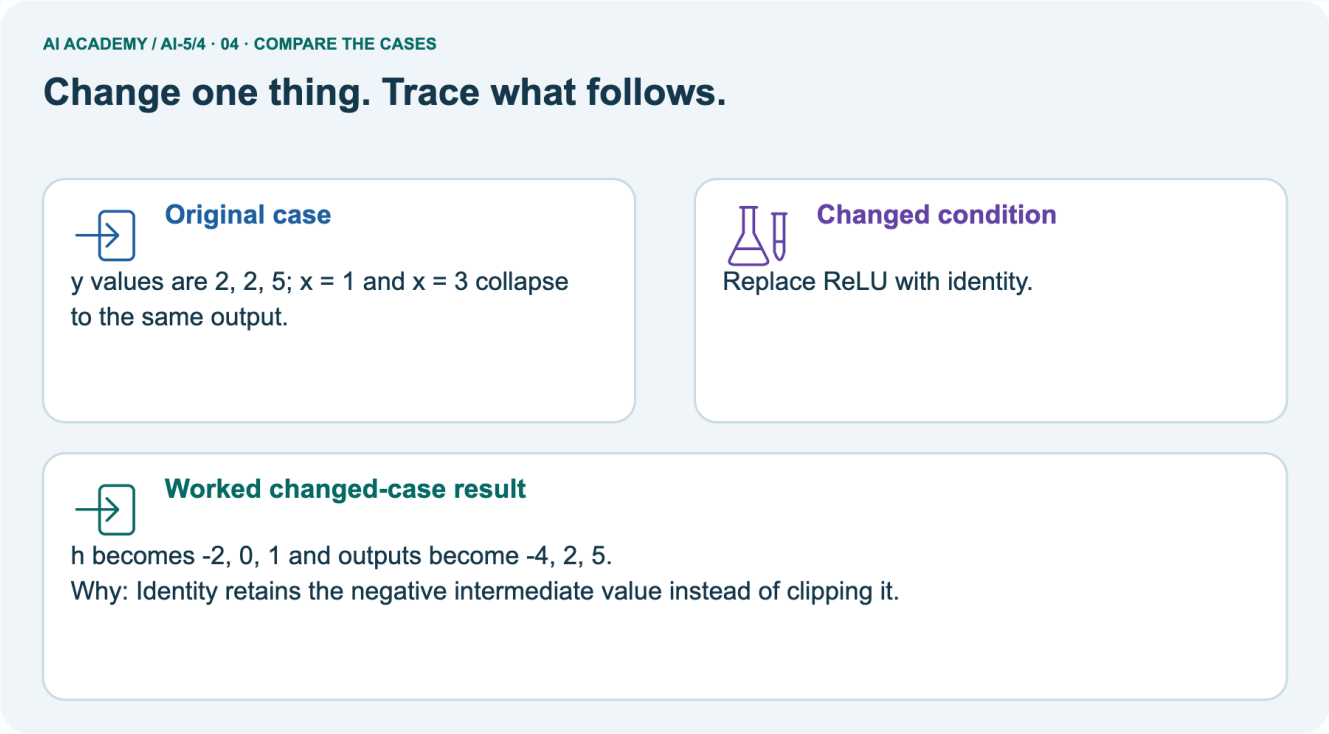
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**. Changed condition: Replace **ReLU** with identity. Expected result: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

Reason: Identity retains the negative intermediate value instead of clipping it.

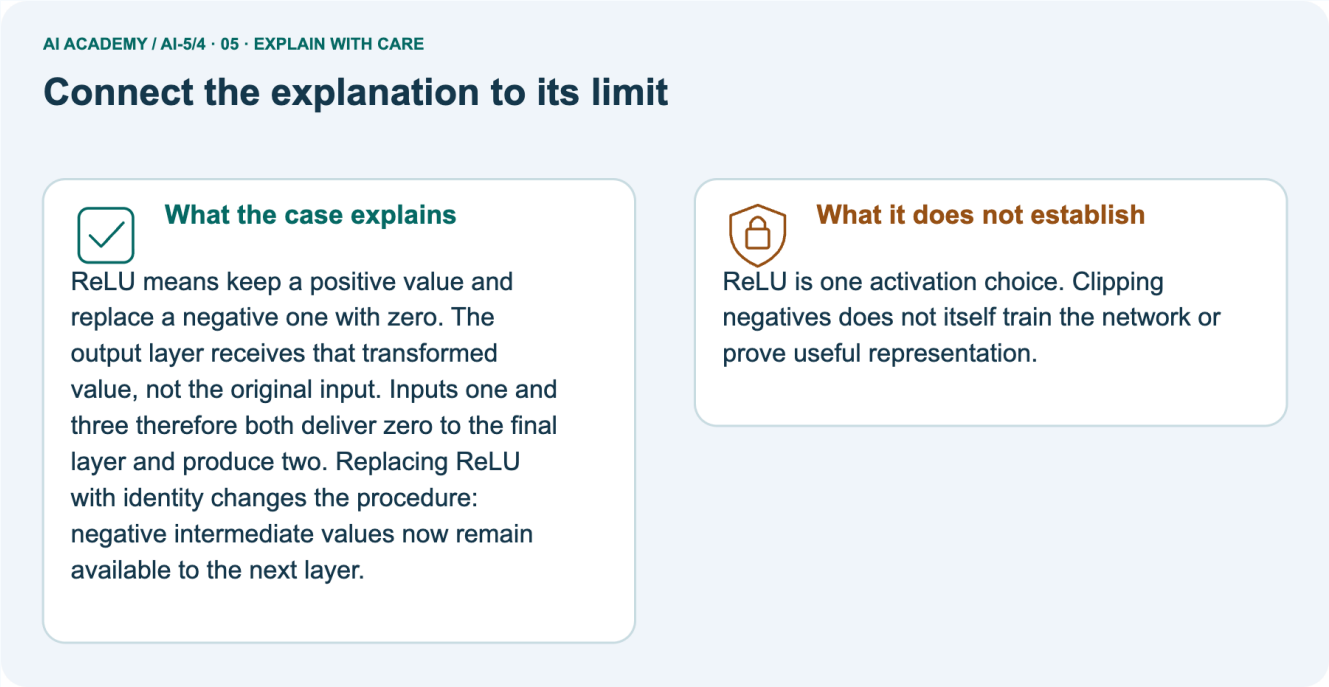
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Identity retains the negative intermediate value instead of clipping it.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

ReLU means keep a positive value and replace a negative one with zero. The **output** layer receives that transformed value, not the original **input**.

Inputs one and three therefore both deliver zero to the final layer and produce two. Replacing **ReLU** with identity changes the procedure: negative intermediate values now remain available to the next layer.

Limit: **ReLU** is one activation choice. Clipping negatives does not itself train the network or prove useful representation.

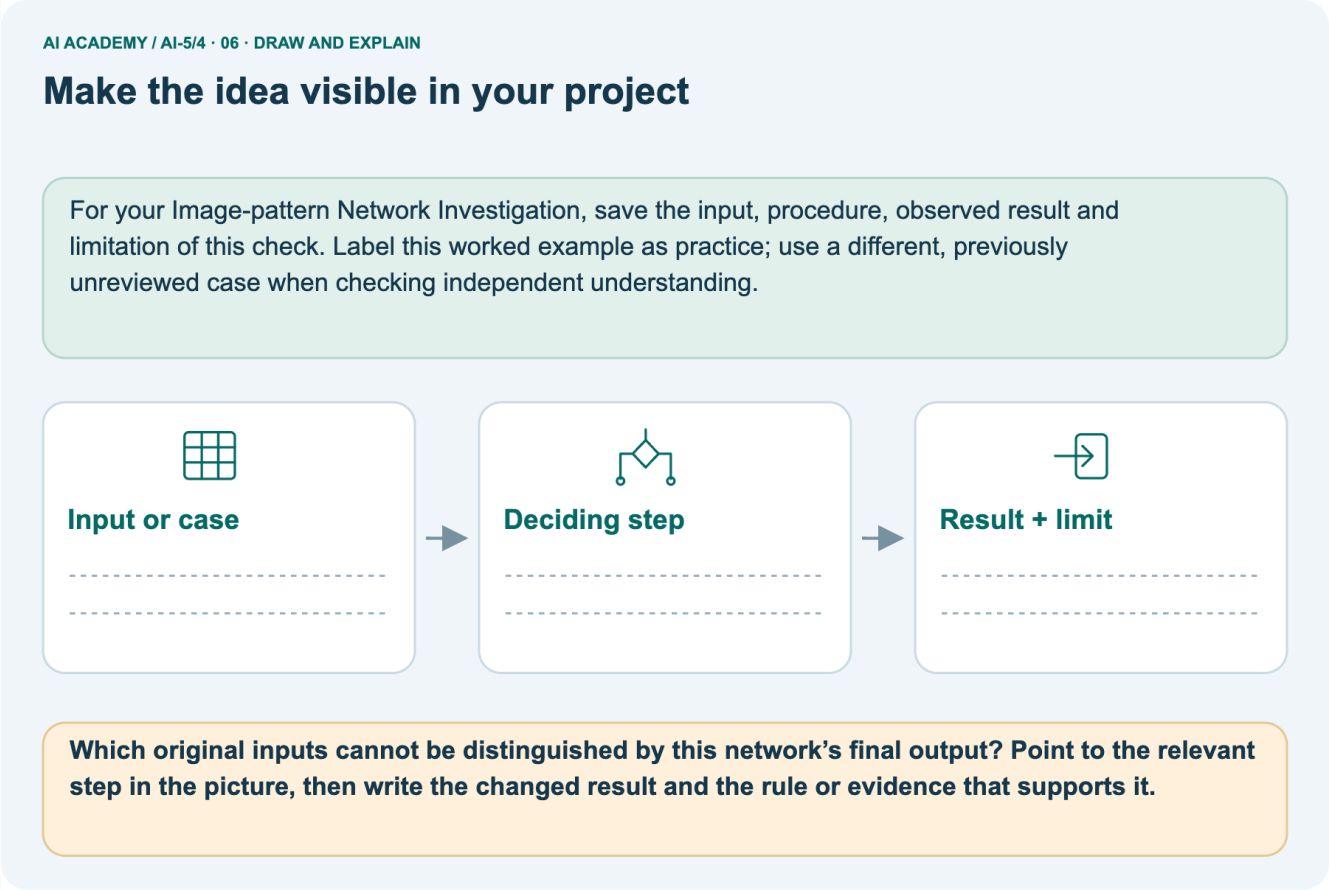
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **ReLU** is one activation choice. Clipping negatives does not itself train the network or prove useful representation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which original **input**s cannot be distinguished by this network's final **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: x=1 and x=3 both give h = 0 and y = 2. After **ReLU**, the downstream step receives identical values for those **input**s. Changed-case check: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

Identity retains the negative intermediate value instead of clipping it.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.